Machine Learning

AIML CZG565/ DSECLZG565

M4 : Linear Models for Classification

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Contact Session 5,6

Decision Theory
&
Objective of Classification Models

Work Integrated Learning Programmes

Inductive Learning Hypothesis : Interpretation

- Target Concept : **t**
- Discrete : f(x) ∈ {Yes, No, Maybe} — Classification
- Continuous : f(x) ∈ [20-100] — Regression
- Probability Estimation : f(x) ∈ [0-1]

Sky	AirTemp	Humidity	Wind	Water	Forecast	*EnjoySport?*
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

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Decision Theory

- Target Concept : **t**
- Discrete : f(x) ∈ {Yes, No} ie., **t** ∈ {0, 1} Binary Classification
- Continuous : f(x) ∈ [20-100]
- Probability Estimation : f(x) ∈ [0-1]

ML Task : Predict the Employability of interview candidates based on CGPA & IQ

Preprocess Implemented:

Min-Max Normalization on IQ

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

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Decision Theory :

The decision problem: given x, predict t according to a probabilistic model p(x, t)

- Target Concept : **t**
- Discrete : f(x) ∈ {Yes, No} ie., **t** ∈ {0, 1}
- Continuous : f(x) ∈ [20-100]
- Probability Estimation : f(x) ∈ [0-1]

$p(x, C_k)$ is the (central!) inference problem

CGPA	IQ	IQ	Job Offered	P(Job = 1)
5.5	6.7	100	1	
X =< 5 ,	7 , >	105	0	0.8 ? = $P(C_k \mid X)$
8	6	90	1	0.4
9	7	105	1	0.75
6	8	120	0	0.95
7.5	7.3	110	0	0.35
				0.4

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Classification Problem: Stages

$$p(\mathcal{C}_k|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\mathbf{x})}.$$

$$= \frac{p(x, C_k)}{p(x)} = \frac{p(x, C_k)}{\sum_{k=1}^{2} p(x, C_k)}$$

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0

Training Set

Learning algorithm

Induction/ Inference step

Learn Model for $p(x,C_k)$

$p(x,C_{job=1})$ & $p(x,C_{job=0})$

Model

Apply Model to find optimal t

Deduction/ Decision Step

CPGA	IQ	Job - Offered
3	4	?
7	6	?
5.5	8	?

Test Set

Decision Region

Sample Rule / Hypothesis:
IF CGPA<7 Job = 1
Else Job = 0

Training Set

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0

Learning algorithm

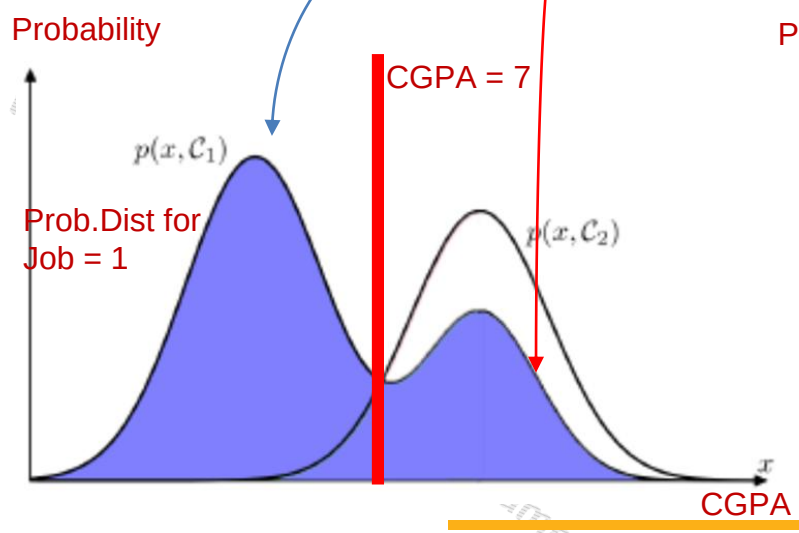
Induction/ Inference step

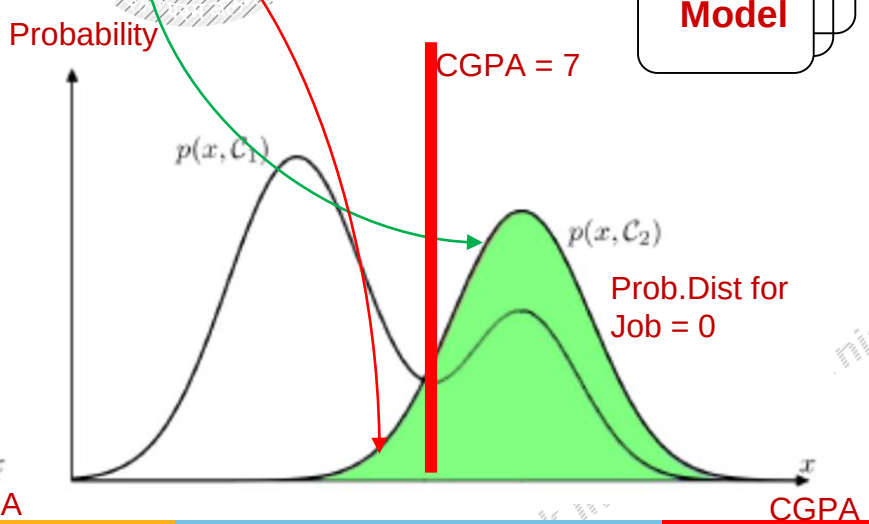
Learn Model for $p(x,C_k)$

Model divides the input space into regions $\mathbf{R}_k$ called **decision regions**, one for each class, such that all points in R_k are assigned to class C_k

A mistake occurs when an input vector belonging to class C_1 is assigned to class C_2

Model





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Misclassification Rate

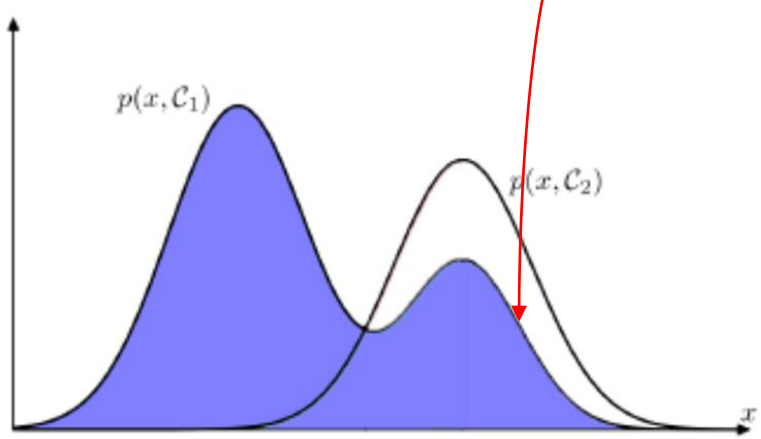
$$p(C_k|x) = \frac{p(x, C_k)}{p(x)}$$

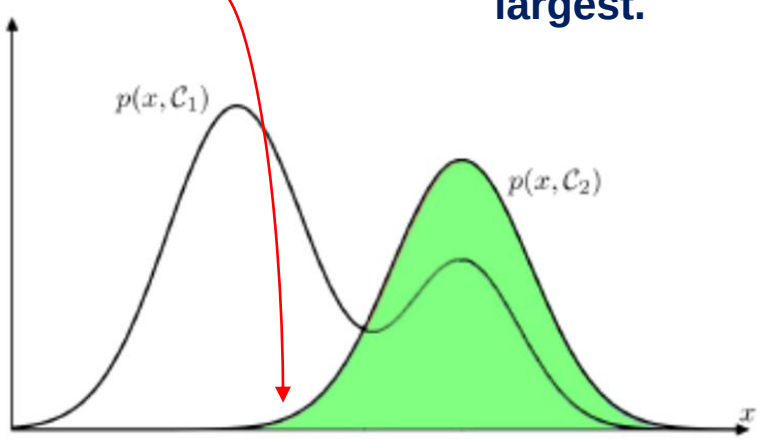
$$\begin{aligned} p(\text{mistake}) &= p(\mathbf{x} \in \mathcal{R}_1, \mathcal{C}_2) + p(\mathbf{x} \in \mathcal{R}_2, \mathcal{C}_1) \\ &= \int_{\mathcal{R}_1} p(\mathbf{x}, \mathcal{C}_2)\,\mathrm{d}\mathbf{x} + \int_{\mathcal{R}_2} p(\mathbf{x}, \mathcal{C}_1)\,\mathrm{d}\mathbf{x}. \end{aligned}$$

To minimize p(mistake), each **x** is assigned to whichever class has the smaller value of the integrand

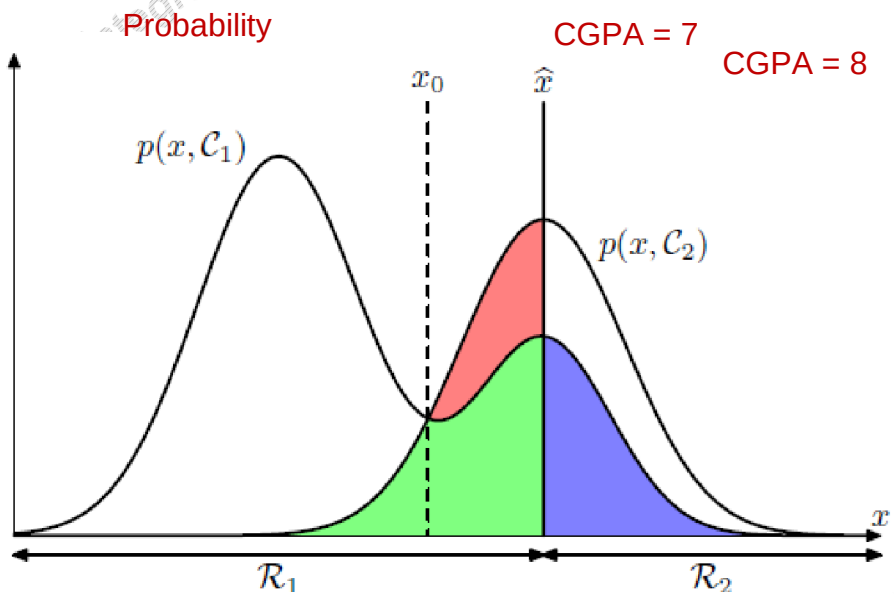
$$\begin{aligned} & p(x, C_1) > p(x, C_2) \\ & \Leftrightarrow p(C_1|x)p(x) > p(C_2|x)p(x) \\ & \Leftrightarrow p(C_1|x) > p(C_2|x) \end{aligned}$$

The minimum probability of making a mistake is obtained if each value of **x** is assigned to the class for which the **posterior probability $p(C_k|x)$ is largest.**





Decision Theory - Summary

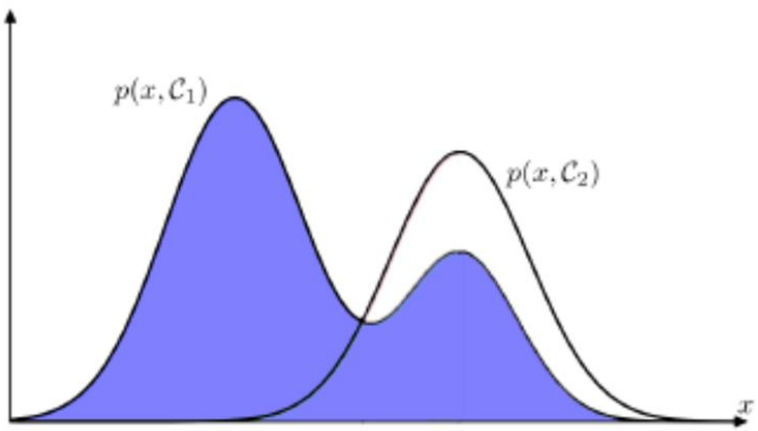


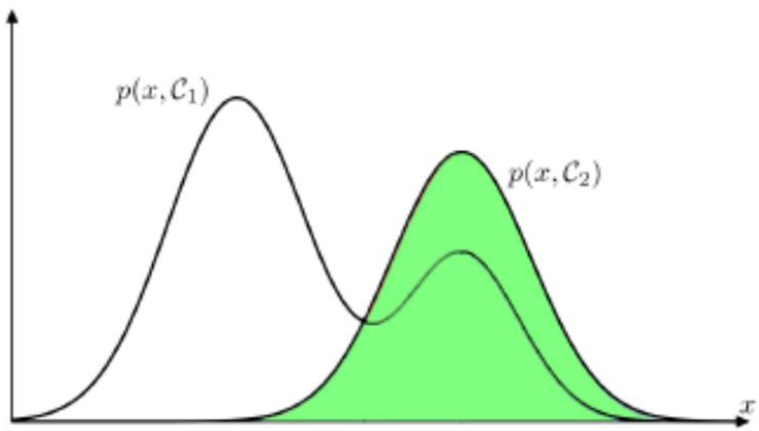
$\hat{x}$: decision boundary.

x_0: optimal decision boundary

$$x_0 : \arg\min_{\mathcal{R}_1}\{p\,(\text{mistake})\}$$

CGPA





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Linear Models for Classification

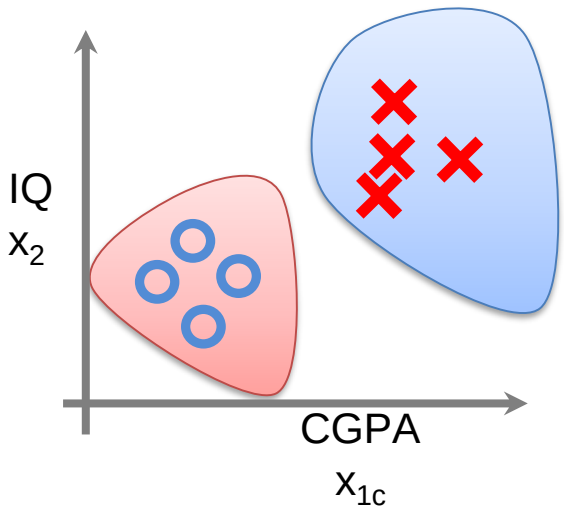
Work Integrated Learning Programmes

Types of Classification

Decision Theory: Interpretation

Model Building

Generative



Likelihood

Class Prior Probability

$$P(c \mid x) = \frac{P(x \mid c)P(c)}{P(x)}$$

Posterior Probability

Predictor Prior Probability

$$P(c \mid X) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...	....	.....

$$P(Y \mid X_1 X_2 \ldots X_n) = \frac{P(X_1 X_2 \ldots X_d \mid Y)P(Y)}{P(X_1 X_2 \ldots X_d)}$$

Known as generative models, because by sampling from them it is possible to generate synthetic data points in the input space.

Eg.,Classification: **Naïve Bayes,**

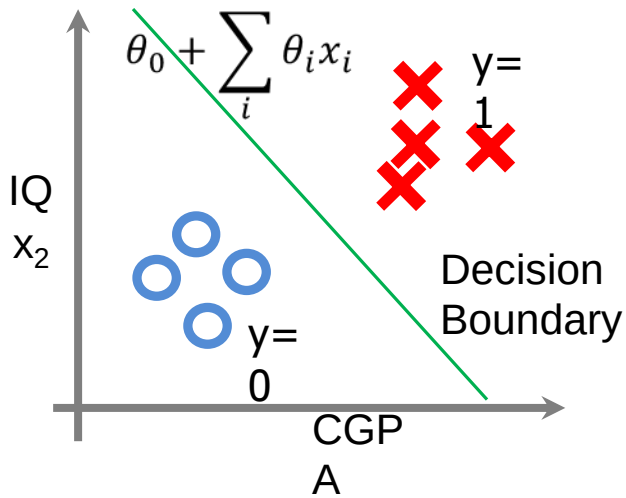
Clustering : **Mixtures of Gaussians**

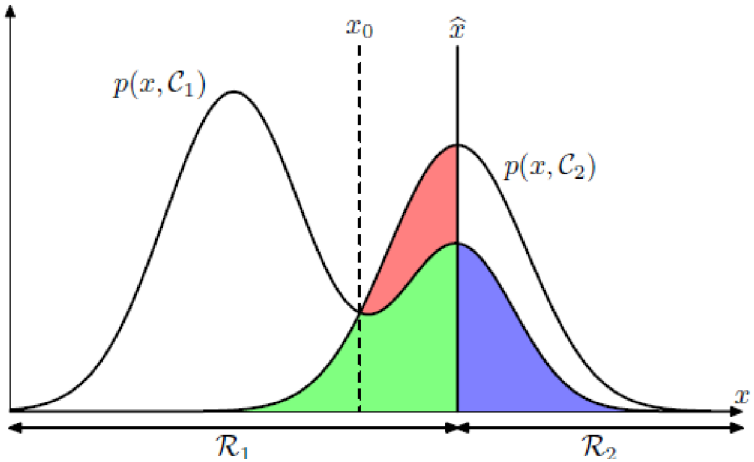
Types of Classification

Decision Theory: Interpretation

Model Building

Discriminative





$$\theta_0 + \sum_i \theta_i x_i \geq 0$$

$$\theta_0 + \sum_i \theta_i x_i < 0$$

$P(c \mid x)$:

Posterior Probability

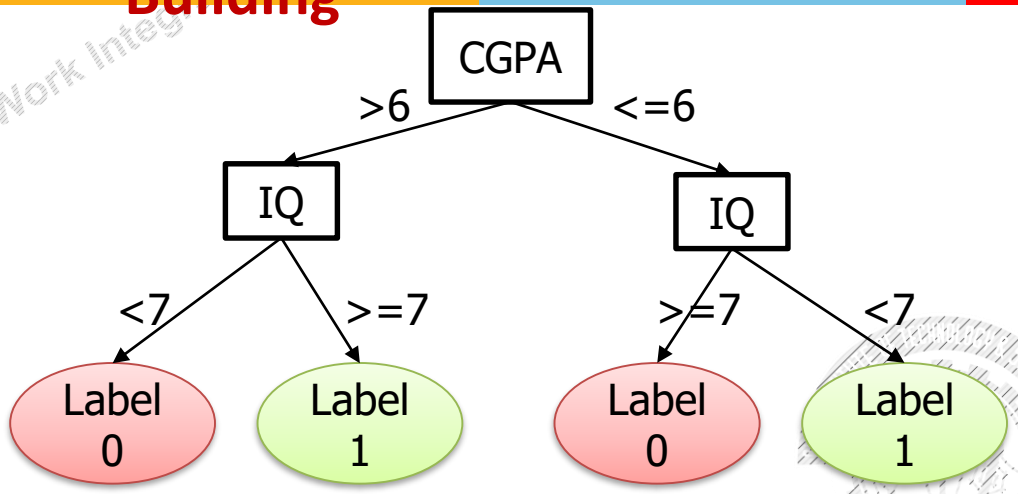
CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...	....	.....

Logistic regression, SVMs , tree based classifiers (e.g. decision tree) Traditional neural networks, **Nearest neighbor**

Types of Classification

Decision Theory: Interpretation

Building **Model**



IF CGPA > 6 and IQ ≥ 7
Job offered = 1
Else If CGPA <=6 and IQ <7
Job offered = 1
Else if CGPA <=6 and IQ >=7
Job offered = 0
.....

Logistic regression, SVMs , tree based classifiers (e.g. decision tree) Traditional neural networks, **Nearest neighbor**

CPGA	IQ	Job - Offered
5.5	6.7	1
5	7	0
8	6	1
9	7	1
6	8	0
7.5	7.3	0
...	....	.....

Logistic Regression

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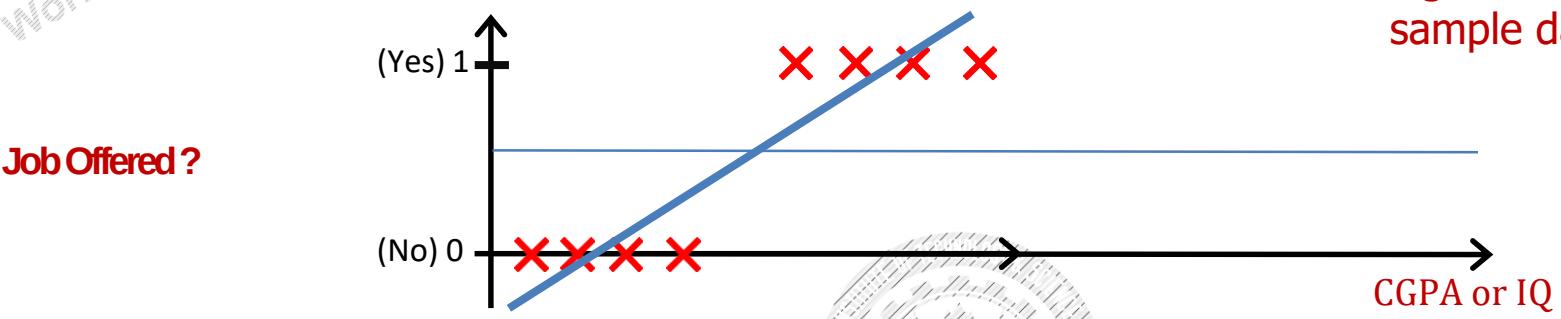
Logistic Regression vs Least Squares Regression

Figure depicts sample data points



- Independent Attribute : CGPA or IQ
- Can we solve the problem using linear

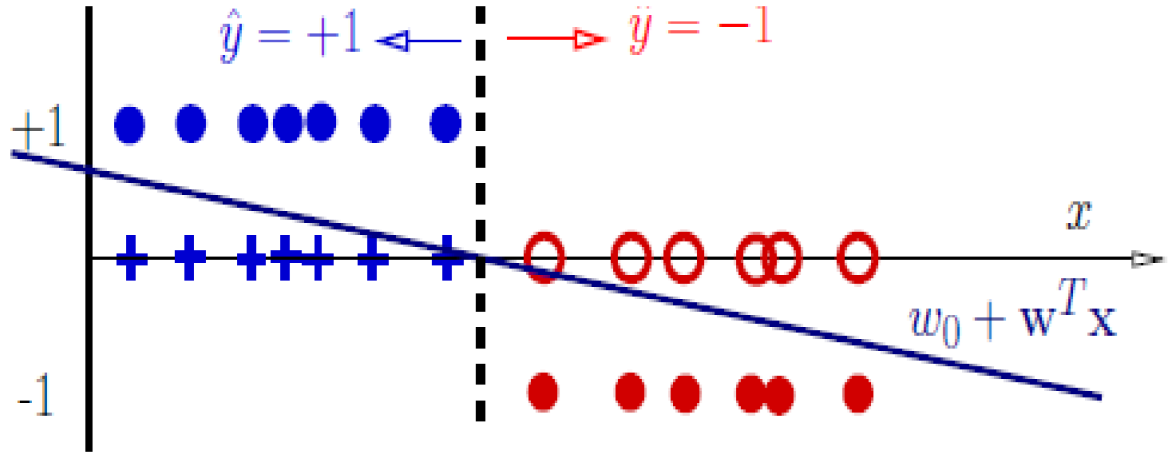
A Discriminant function f (x) directly map input to class labels
In two-class problem, f (.) is binary valued

If $h_\theta(x) \geq 0.5$, predict "y = 1"

If $h_\theta(x) < 0.5$, predict "y = 0"

In this use case : $h_\theta(x) = 0.7$, implies that there is 70% of chance of the candidate being selected in the interview

Decision Rules



- Classifier:

$$f(\mathbf{x}, \mathbf{w}) = w_o + \mathbf{w}^T\mathbf{x} \quad \text{(linear discriminant fur}$$

$$y = \begin{cases} 1 & \text{if } f(\mathbf{x}, \mathbf{w}) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

- Decision rule is

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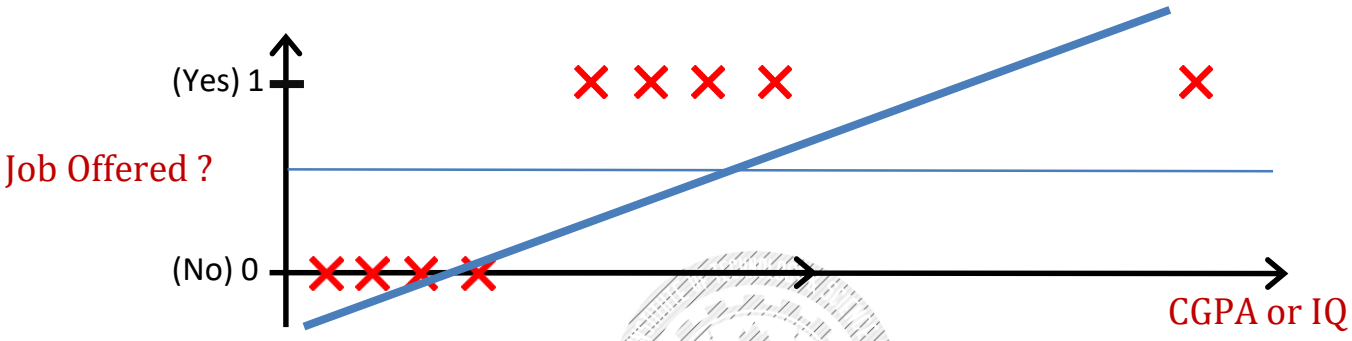
- Mathematically

$$y = \text{sign}(w_0 + \mathbf{w}^T\mathbf{x})$$

Logistic Regression vs Least Squares Regression



Failure due to adding a new point

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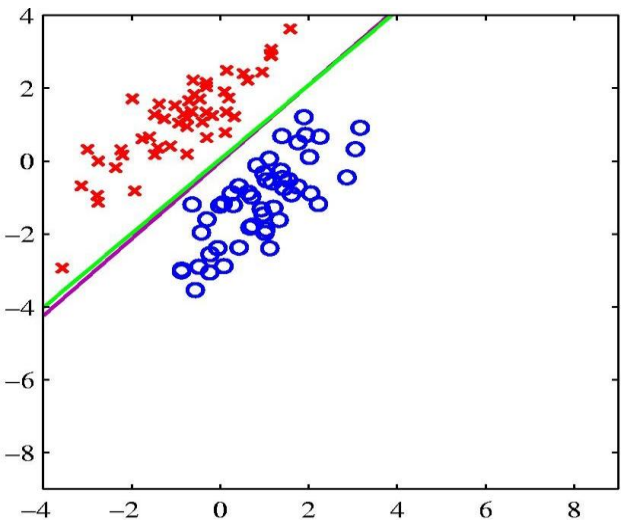
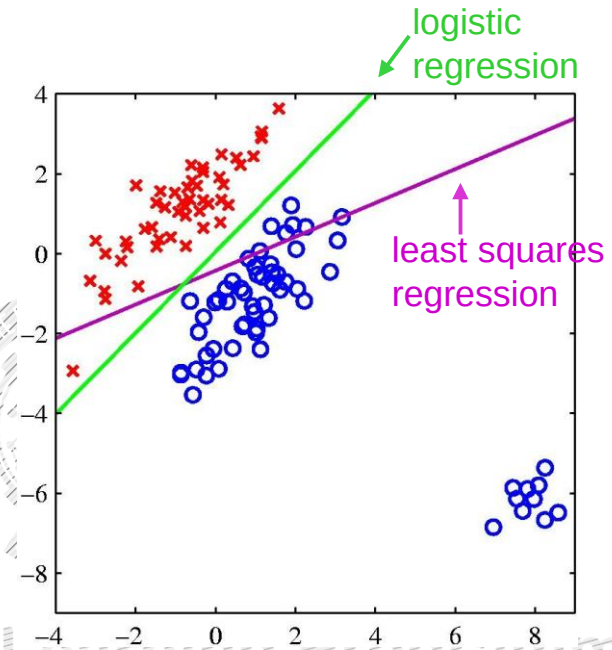
Logistic Regression vs Least Squares Regression





The right-hand plot shows the corresponding results obtained when extra data points are added at the bottom left of the diagram, showing that least squares is highly sensitive to outliers, unlike logistic regression.

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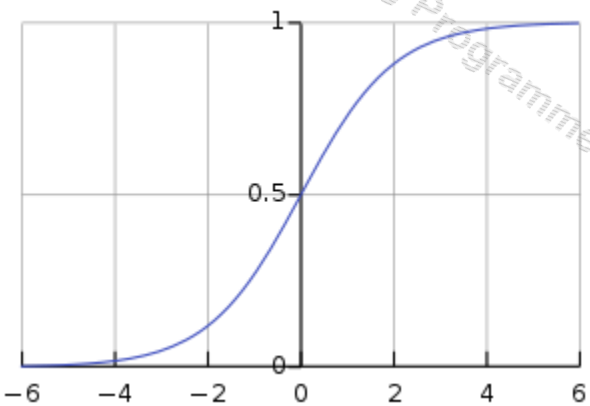
Sigmoid Function

- Sigmoid/logistic function takes a real value as input and outputs another value between 0 and 1
- That framework is called logistic regression
  - Logistic: A special mathematical sigmoid function it uses
  - Regression: Combines a weight vector with observations to create an answer

- Want $0 \leq h_\theta(x) \leq 1$

- $h_\theta(x) = g(\theta^\top x)$,

where $g(z) = \frac{1}{1+e^{-z}}$



$$h_\theta(x) = g(\theta^T x)$$

- Sigmoid function
- Logistic function

$$\boxed{h_\theta(x) = \frac{1}{1+e^{-\theta^\top x}}}$$

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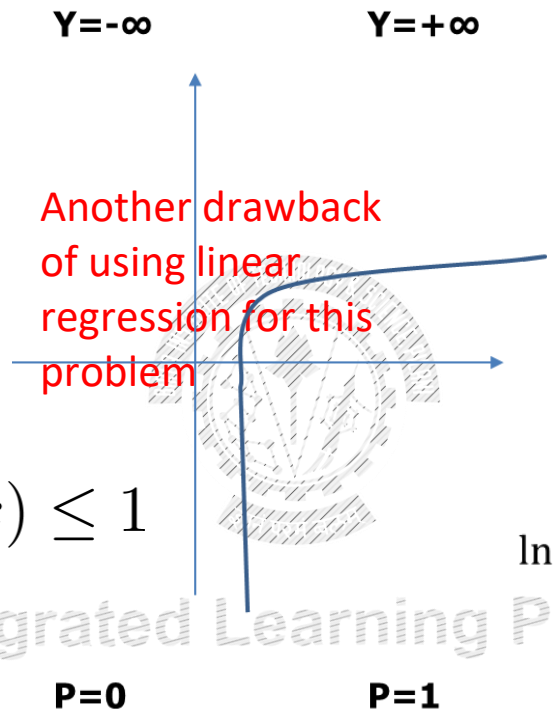
Logistic Regression vs Least Squares Regression



Another drawback of using linear regression for this problem

$$0 \leq h_\theta(x) \leq 1$$

$$Y = \theta_0 + \sum_i \theta_i x_i$$

$$p = \frac{1}{1 + e^{-y}}$$

$$h_\theta(x) = g(\theta^T x)$$

$$\ln \frac{p}{1-p} = \theta_0 + \sum_i \theta_i x_i$$

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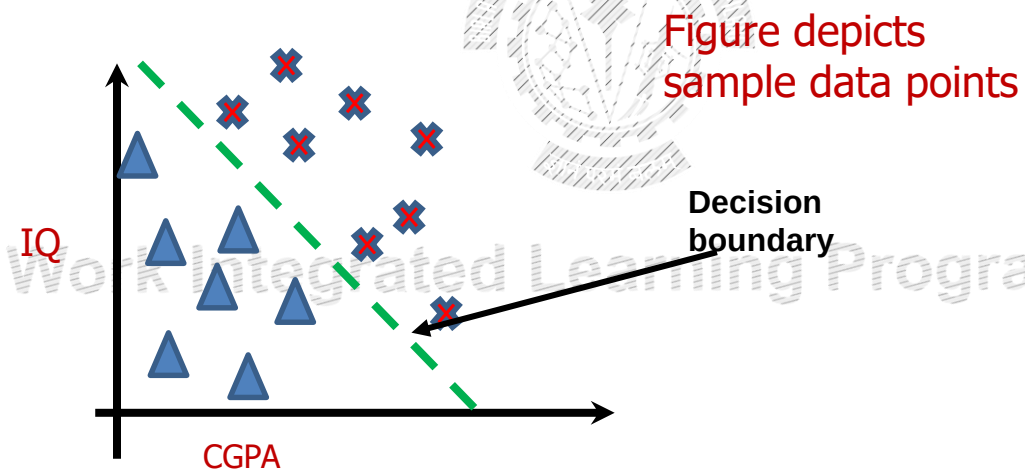
Logistic Regression – Sample Linear Boundary

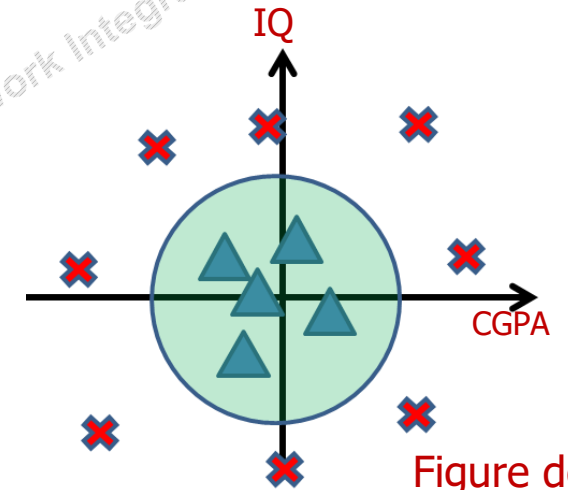
- At decision boundary output of logistic regression is 0.5
- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$
  - e.g., $\theta_0 = -3,\ \theta_1 = 1,\ \theta_2 = 1$



- Predict "$y = 1$" if $-3 + x_1 + x_2 \geq 0$

Slide credit: Andrew Ng

Logistic Regression – Sample Non-Linear Boundary

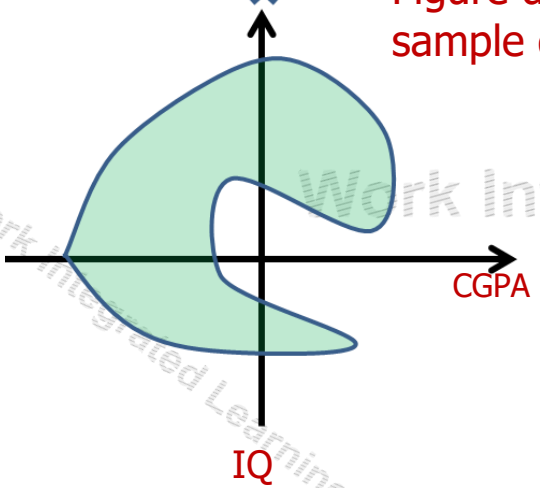


- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$

E.g., $\theta_0 = -1, \theta_1 = 0, \theta_2 = 0, \theta_3 = 1, \theta_4 = 1$

- Predict "$y = 1$" if $-1 + x_1^2 + x_2^2 \geq 0$

Figure depicts sample data points



- $h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \cdots)$

Slide credit: Andrew Ng

Learning model parameters

- Training set:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$$

- m examples

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \cdots \\ x_n \end{bmatrix} \qquad x_0 = 1, y \in \{0, 1\}$$

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

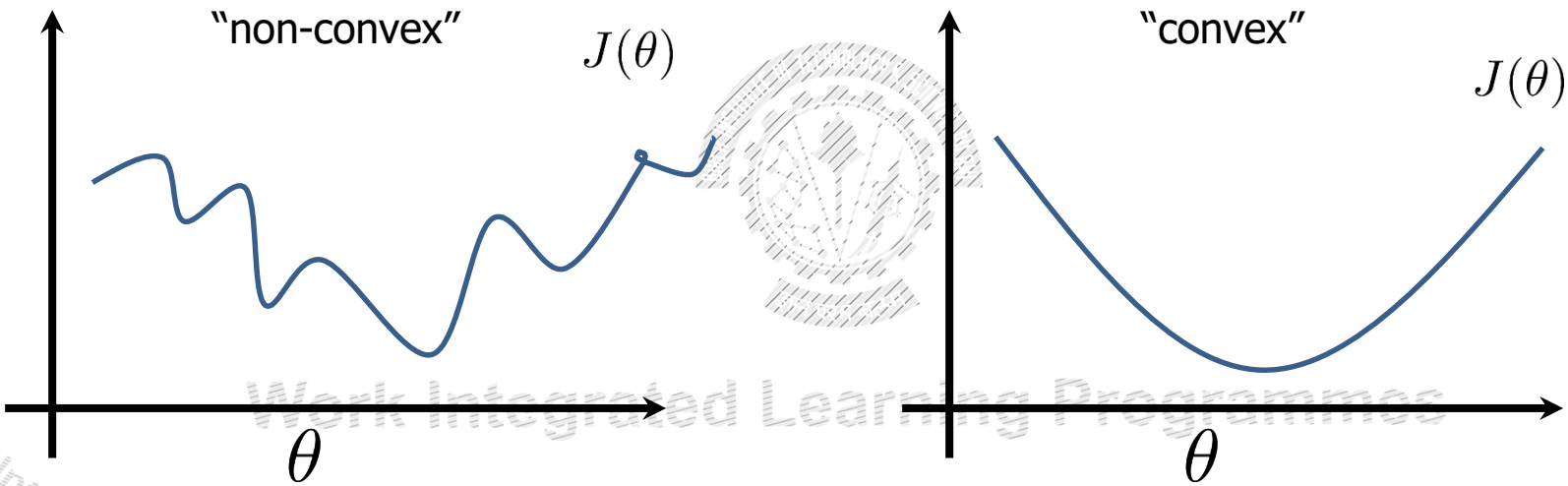
- **How to choose parameters (feature weights) ?**

Notion of Cost Function in Classification

Work Integrated Learning Programmes

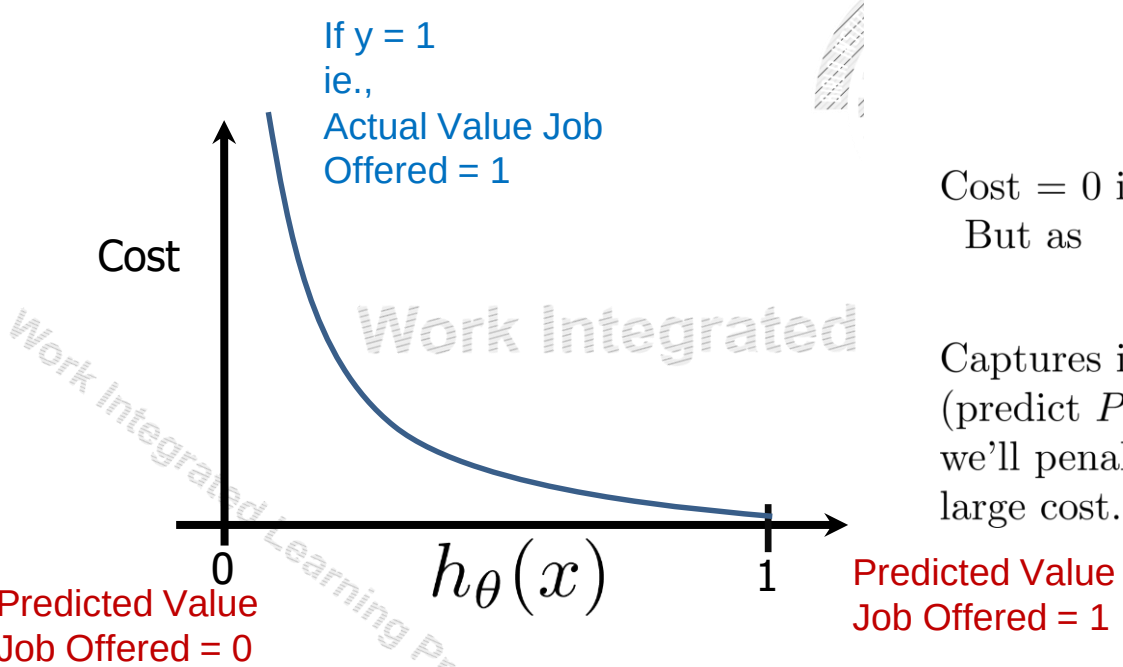
Logistic Regression

- Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \cdots, (x^{(m)}, y^{(m)})\}$
- How to choose parameters (feature weights)? θ



Logistic regression cost function (cross entropy)

$$\mathrm{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$



$\mathrm{Cost} = 0$ if $y = 1, h_\theta(x) = 1$

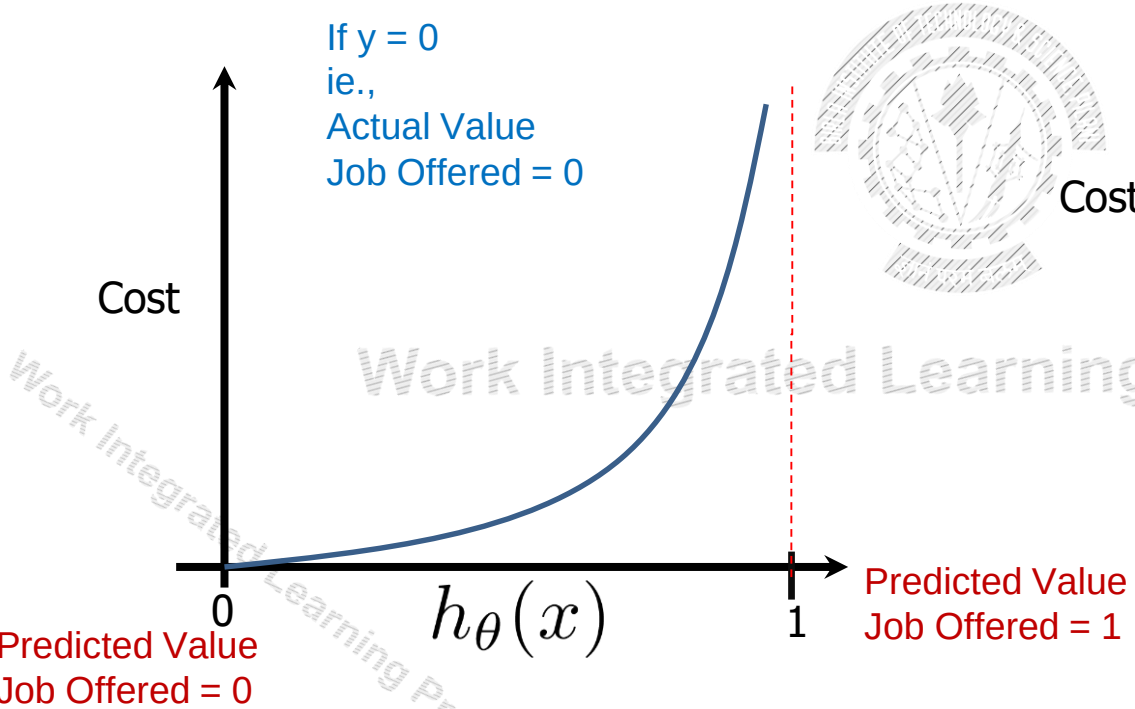
But as $h_\theta(x) \to 0$

$Cost \to \infty$

Captures intuition that if $h_\theta(x) = 0$, (predict $P(y = 1|x; \theta) = 0$), but $y = 1$, we'll penalize learning algorithm by a very large cost.

Logistic regression cost function

$$\text{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$



Cost=0; If y=0 and $h_\theta(x)=0$

Work Integrated Learning Programmes

Cost function

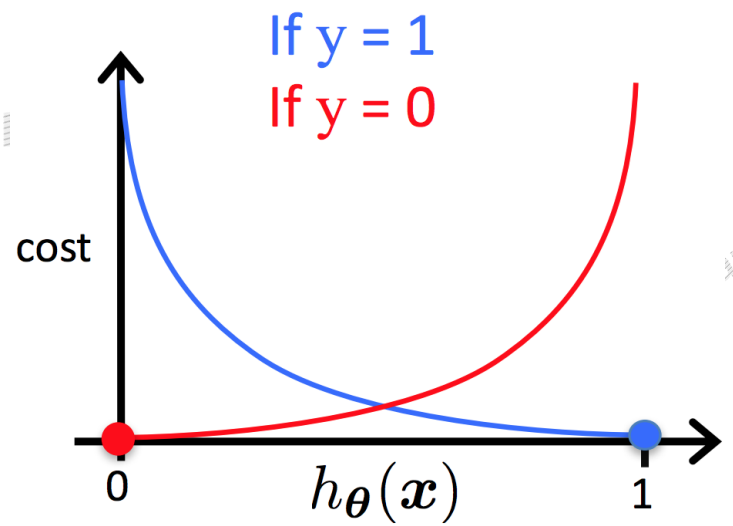
$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

$$= -\frac{1}{m}[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1-y^{(i)}) \log (1-h_\theta(x^{(i)}))]$$

To fit parameters θ : Apply Gradient Descent Algorithm $\min_\theta J(\theta)$

To make a prediction given new x :

Output : $h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$



Gradient Descent Algorithm

$$J(\theta) = -\frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \log\left(h_\theta\left(x^{(i)}\right)\right) + (1 - y^{(i)}) \log\left(1 - h_\theta\left(x^{(i)}\right)\right)\right]$$

Goal: $\min_{\theta} J(\theta)$

Repeat

{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

}

$$\boxed{\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right) x_j^{(i)}}$$

Work Integrated Learning Programmes

Gradient Descent Algorithm

Linear Regression

- Repeat {

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right) x_j^{(i)}$$

}

$$h_\theta(x) = \theta^\top x$$

Logistic Regression

Repeat {

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right) x_j^{(i)}$$

}

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^\top x}}$$

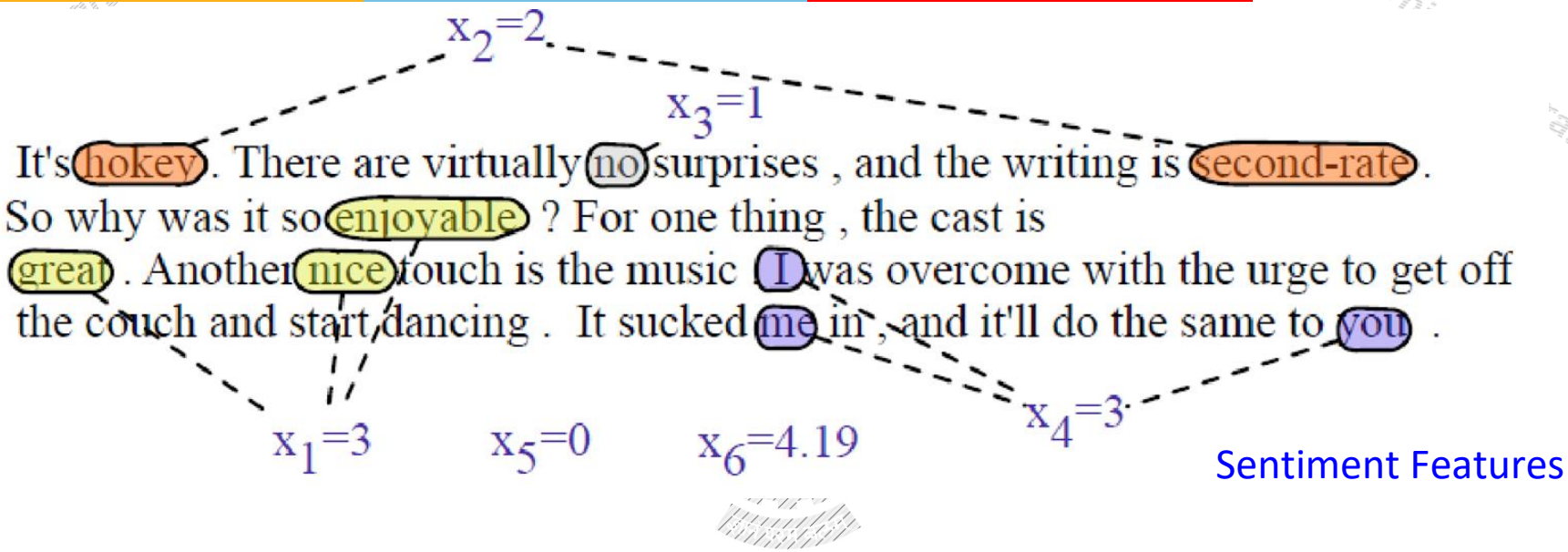
$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 CGPA + \theta_2 IQ)}}$$

Slide credit: Andrew Ng

Application of Logistic Regression
&
Problem Types

Work Integrated Learning Programmes

Example: Sentiment Analysis – With Engineered features



Work Integrated Learning Programmes

Example: Sentiment Analysis – With Engineered features

$x_2=2$ $x_3=1$

It's hokey . There are virtually no surprises , and the writing is second-rate .
So why was it so enjoyable ? For one thing , the cast is
great . Another nice touch is the music . I was overcome with the urge to get off
the couch and start dancing . It sucked me in , and it'll do the same to you .

$x_1=3$ $x_5=0$ $x_6=4.19$ $x_4=3$

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(66) = 4.19$

Sentiment Features

Work Integrated Learning Programmes

Classifying sentiment using logistic regression

Suppose $w = [2.5, -5.0, -1.2, 0.5, 2.0, 0.7]$

$b = 0.1$

$$
\begin{aligned}
p(+|x) = P(Y=1|x) &= \sigma(w \cdot x + b) \\
&= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.19] + 0.1) \\
&= \sigma(.833) \\
&= 0.70
\end{aligned}
$$

$$
\begin{aligned}
p(-|x) = P(Y=0|x) &= 1 - \sigma(w \cdot x + b) \\
&= 0.30
\end{aligned}
$$

Logistic Regression – Fit a Model – Apply Gradient Descent

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

Hyper parameters:
Learning Rate = 0.3
Initial Weights = (0.5, 0.5,0.5)
Regularization Constant = 0

$\theta^{T}X$ = 0.5 +0.5 CGPA + 0.5 IQ

$$h_\theta(x) = \frac{1}{1+e^{-\theta^{T}x}}$$

$$h_\theta(x) = \frac{1}{1+e^{-(0.5\,+0.5\,CGPA\,+\,0.5\,IQ\,)}}$$

$$\theta_0 := \theta_0 - 0.3\frac{1}{6}\sum_{i=1}^{6}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)(1)$$

$$\theta_{CGPA} := \theta_{CGPA} - 0.3\frac{1}{6}\sum_{i=1}^{6}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_{CGPA}^{(i)}$$

$$\theta_{IQ} := \theta_{IQ} - 0.3\frac{1}{6}\sum_{i=1}^{6}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_{IQ}^{(i)}$$

Approx. : New weights

θ_0 = 0.4

$\theta_{1=CGPA}$ =-0.4

$\theta_{2=IQ}$ $= -0.6$

Logistic Regression – Inference & Interpretation

CGPA	IQ	IQ	Job Offered
5.5	6.7	100	1
5	7	105	0
8	6	90	1
9	7	105	1
6	8	120	0
7.5	7.3	110	0

Assume : 0.4+0.3CGPA-0.45IQ

Predict the Job offered for a candidate : (5, 6)
h(x) = 0.31
Y-Predicted = 0 / No

Note :
The exponential function of the regression coefficient ($e^{w\text{-}cpga}$) is the odds ratio associated with a one-unit increase in the cgpa.
+ The odd of being offered with job increase by a factor of 1.35 for every unit increase in the CGPA [np.exp(model.params)]

Regularization

Note : This topic is already covered in the module 3 and the implementation remains the same.

Only one points added here w.r.t interpretation for logistic regression

Ways to Control Overfitting – Interpretation of Hyper parameter

- Regularization

$$Loss(S) = \sum_{i}^{n} Loss(\hat{y_i}, y_i) \quad + \alpha \sum_{j}^{\#Weights} |\theta_j|$$

Note:
The hyperparameter controlling the regularization strength of a Scikit-Learn LogisticRegression model is not alpha (as in other linear models), but its inverse: C. The higher the value of C, the less the model is regularized.

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Evaluation of Classifiers

Using Another Example

Evaluation of Classification Model

Confusion Matrix

Mileage (in kmpl)	*Car Price (in cr)*
9.8	High
9.12	Low
9.5	High
10 ….	Low …

		PREDICTED CLASS	
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	a (TP)	b (FN)
	Class= High	c (FP)	d (TN)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Unseen Data	
Mileage (in kmpl)	***Car Price (in cr)***
7.5	High
10	Low
….	…..

Most effective when the class distribution is relatively balanced

Model 1

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5\,\text{Mileage} - 1.5\,\text{Mileage}^2}}$$

Accuracy: 99%

Model 2

$$\text{CarPrice} = \frac{1}{1+e^{5.5-1.5\,\text{Mileage}}}$$

Accuracy: 50%

Evaluation of Classification Model

Confusion Matrix

	PREDICTED CLASS		
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	0 (TP)	10 (FN)
	Class= High	0 (FP)	990 (TN)

	PREDICTED CLASS		
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	10 (TP)	0 (FN)
	Class= High	500 (FP)	490 (TN)

If a model predicts everything to be class NO, accuracy is 990/1000 = 99 %. This is misleading because this trivial model does not detect any class YES example Detecting the rare class is usually more interesting (e.g., frauds, intrusions, defects, etc)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Which model is better?

Model 1

$$\text{CarPrice} = \frac{1}{1+e^{-8.5 + 0.5\,\text{Mileage} - 1.5\,\text{Mileage}^2}}$$

Accuracy: 99%

Model 2

$$\text{CarPrice} = \frac{1}{1+e^{5.5-1.5\,\text{Mileage}}}$$

Accuracy: 50%

Evaluation of Classification Model

Confusion Matrix

		PREDICTED CLASS	
ACTUAL CLASS		Class= Low	Class= High
	Class= Low	a (TP)	b (FN)
	Class= High	c (FP)	d (TN)

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN}$$

$$ErrorRate = 1 - accuracy$$

$$Precision = Positive\ Predictive\ Value = \frac{TP}{TP + FP}$$

$$Recall = Sensitivity = TP\ Rate = \frac{TP}{TP + FN}$$

$$Specificity = TN\ Rate = \frac{TN}{TN + FP}$$

$$FP\ Rate = \alpha = \frac{FP}{TN + FP} = 1 - specificity$$

$$FN\ Rate = \beta = \frac{FN}{FN + TP} = 1 - sensitivity$$

$$Power = sensitivity = 1 - \beta$$

The F-score (also known as the F1 score or F-measure, combines precision and recall into a single score . F1-score is a better metric when there are imbalanced classes (More on this in upcoming slides)

F-score
= 2 * (precision * recall) / (precision + recall)

Which Classifier is better?

Low Skew Case

T1	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	50	50
	Class=No	1	99

$$\text{Precision (p)} = 0.98$$
$$\text{TPR} = \text{Recall (r)} = 0.5$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 50$$
$$\text{F} - \text{measure} = 0.66$$

T2	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	10	90

$$\text{Precision (p)} = 0.9$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.1$$
$$\text{TPR/FPR} = 9.9$$
$$\text{F} - \text{measure} = 0.94$$

T3	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	1	99

$$\text{Precision (p)} = 0.99$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 99$$
$$\text{F} - \text{measure} = 0.99$$

Which Classifier is better?

Medium Skew Case

T1	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	50	50
	Class=No	10	990

$$\text{Precision (p)} = 0.83$$
$$\text{TPR} = \text{Recall (r)} = 0.5$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 50$$
$$\text{F} - \text{measure} = 0.62$$

T2	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	100	900

$$\text{Precision (p)} = 0.5$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.1$$
$$\text{TPR/FPR} = 9.9$$
$$\text{F} - \text{measure} = 0.66$$

T3	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	10	990

$$\text{Precision (p)} = 0.9$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 99$$
$$\text{F} - \text{measure} = 0.94$$

Which Classifier is better?

High Skew Case

T1	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	50	50
	Class=No	100	9900

$$\text{Precision (p)} = 0.3$$
$$\text{TPR} = \text{Recall (r)} = 0.5$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 50$$
$$\text{F} - \text{measure} = 0.375$$

T2	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	1000	9000

$$\text{Precision (p)} = 0.09$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.1$$
$$\text{TPR/FPR} = 9.9$$
$$\text{F} - \text{measure} = 0.165$$

T3	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	99	1
	Class=No	100	9900

$$\text{Precision (p)} = 0.5$$
$$\text{TPR} = \text{Recall (r)} = 0.99$$
$$\text{FPR} = 0.01$$
$$\text{TPR/FPR} = 99$$
$$\text{F} - \text{measure} = 0.66$$

Which Model should you use?

	False Positive Rate	False Negative Rate
Model 1	41%	3%
Model 2	5%	25%

Mistakes have different costs:

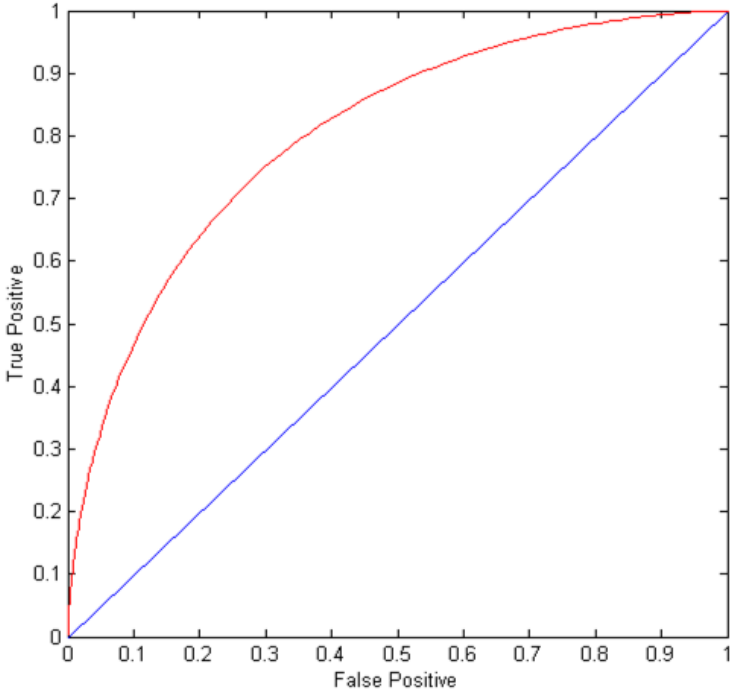
- Disease Screening – LOW FN Rate
- Spam filtering – LOW FP Rate

Conservative vs Aggressive settings:

- The same application might need multiple tradeoffs

ROC (Receiver Operating Characteristic)

- A graphical approach for displaying trade-off between detection rate and false alarm rate
- Developed in 1950s for signal detection theory to analyze noisy signals
- ROC curve plots TPR against FPR
  - Performance of a model represented as a point in an ROC curve

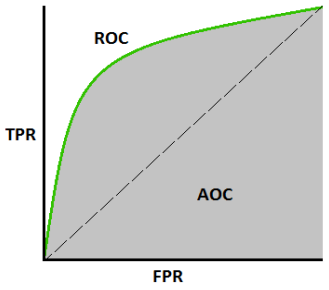


(TPR,FPR):

- (0,0): declare everything to be negative class
- (1,1): declare everything to be positive class
- (1,0): ideal
- Diagonal line:
  - Random guessing
  - Below diagonal line:
    - prediction is opposite of the true class

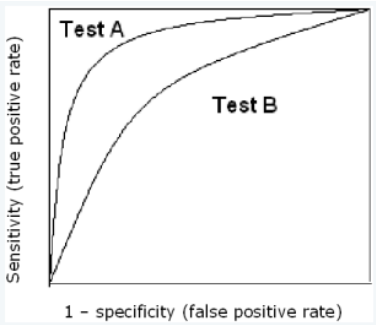
Receiver Operating Characteristic (ROC) Curve

- ROC curve plots TPR and FPR, to graphically represent their trade-off
- AUC represents degree or measure of separability. It tells how much model is capable of distinguishing between classes.
  - Higher the AUC, better the model is at predicting
- Area Under Curve (AUC) of ROC – evaluates model performance on ave
  - AUC of ROC = 1, for a perfect model
  - AUC of ROC = 0.5, if the model is random
- For model comparison, AUC of ROC should be larger for the model to be superior or better performing



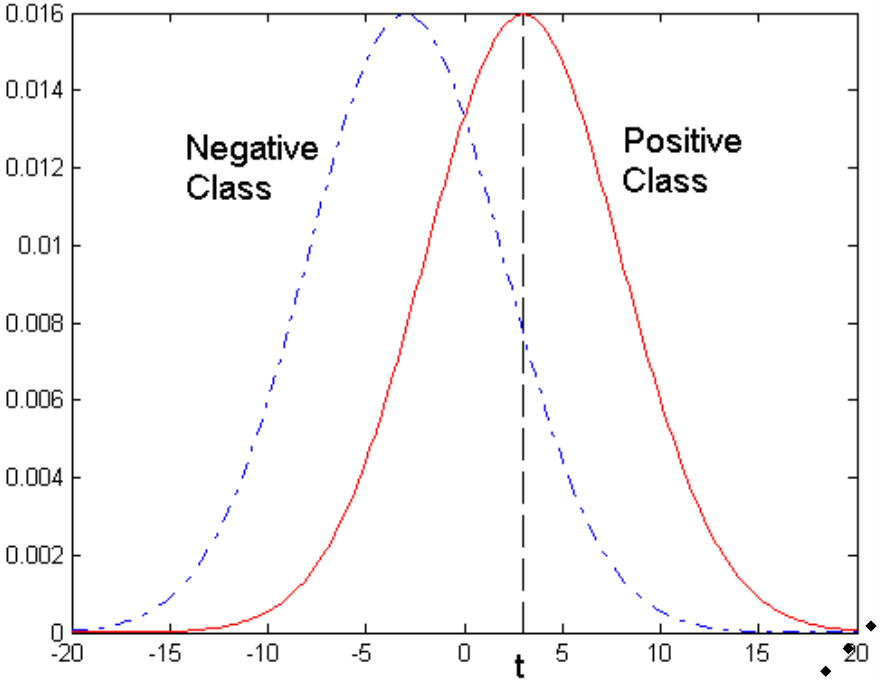
Usage

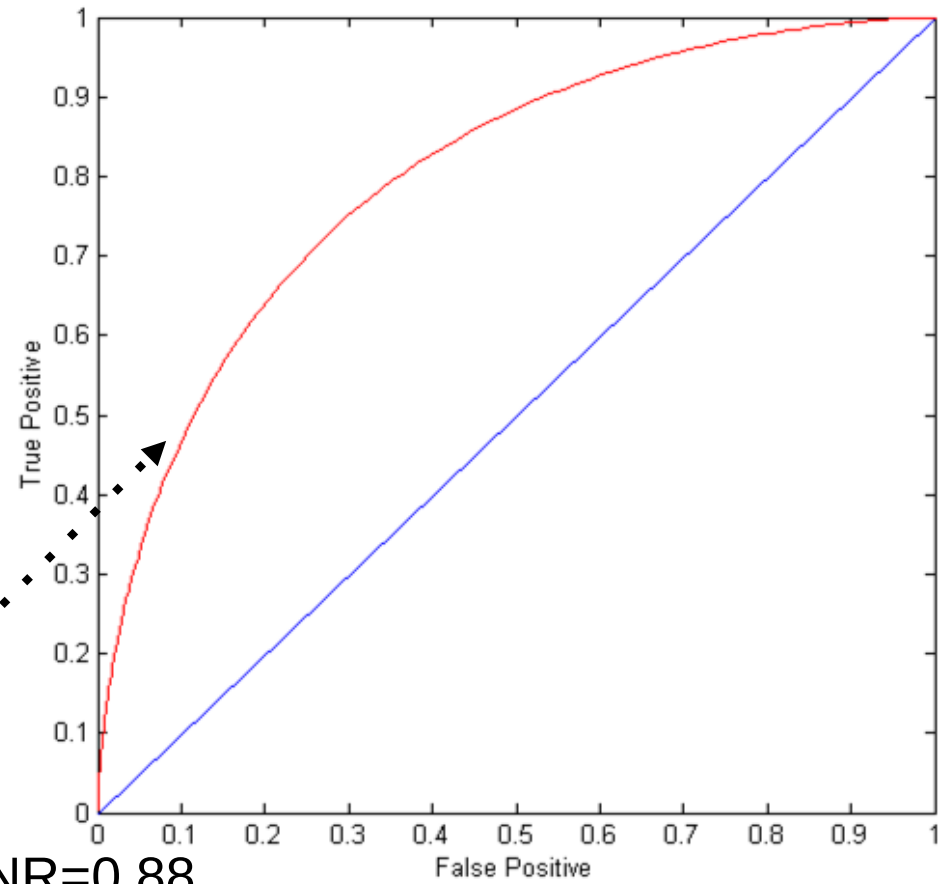
- Threshold selection
- Performance assessment
- Classifier comparison



ROC Curve Example

- 1-dimensional data set containing 2 classes (positive and negative)
- Any points located at x > t is classified as positive





At threshold t:

TPR=0.5, FNR=0.5, FPR=0.12, TNR=0.88

How to Construct an ROC curve

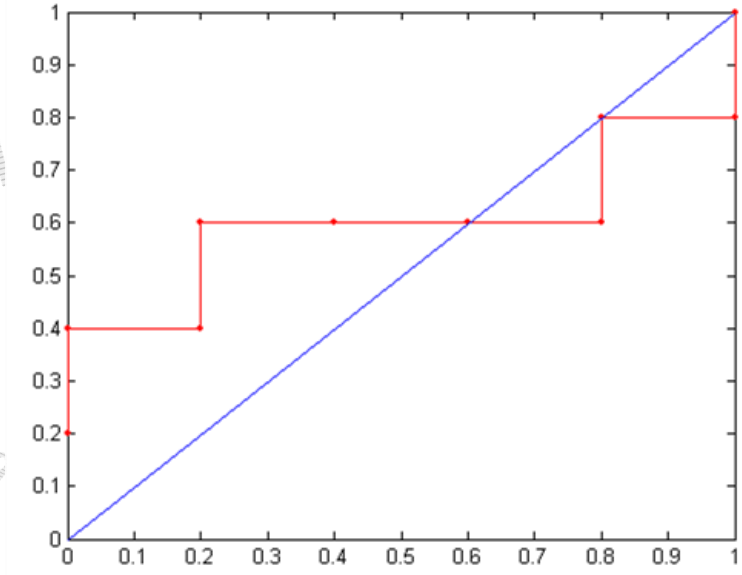
Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- Use a classifier that produces a continuous-valued score for each instance
  - The more likely it is for the instance to be in the + class, the higher the score
- Sort the instances in decreasing order according to the score
- Apply a threshold at each unique value of the score
- Count the number of TP, FP, TN, FN at each threshold
  - TPR = TP/(TP+FN)
  - FPR = FP/(FP + TN)

How to construct an ROC curve

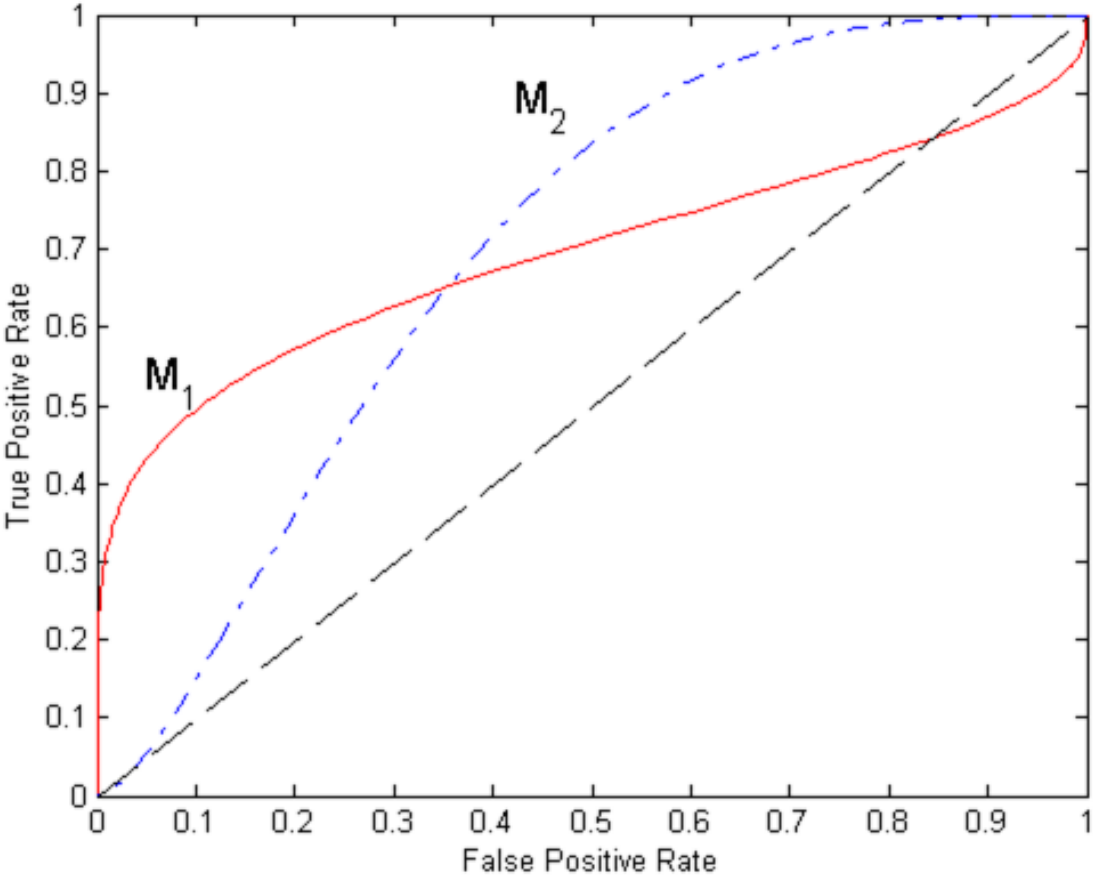
	Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=		0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
	TP	5	4	4	3	3	3	3	2	2	1	0
	FP	5	5	4	4	3	2	1	1	0	0	0
	TN	0	0	1	1	2	3	4	4	5	5	5
	FN	0	1	1	2	2	2	2	3	3	4	5
→	TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
→	FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

ROC Curve:



Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

Using ROC for Model Comparison

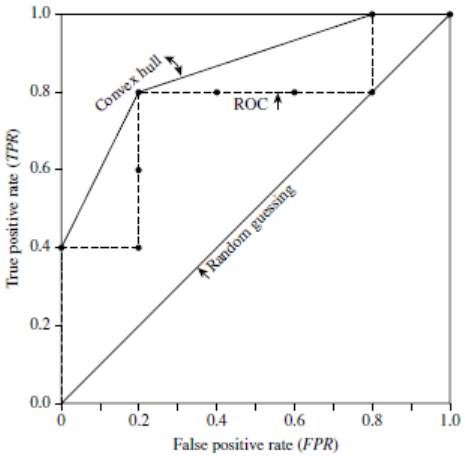


- No model consistently outperforms the other
  - M_1 is better for small FPR
  - M_2 is better for large FPR
- Area Under the ROC curve (AUC)
  - Ideal:
    - Area = 1
  - Random guess:
    - Area = 0.5

Example

The table below shows the probability value (column 3) returned by a probabilistic classifier for each of the 10 tuples in a test set, sorted by decreasing probability order. The corresponding ROC is given on right hand side.

Tuple #	Class	Prob.	TP	FP	TN	FN	TPR	FPR
1	*P*	0.90	1	0	5	4	0.2	0
2	*P*	0.80	2	0	5	3	0.4	0
3	*N*	0.70	2	1	4	3	0.4	0.2
4	*P*	0.60	3	1	4	2	0.6	0.2
5	*P*	0.55	4	1	4	1	0.8	0.2
6	*N*	0.54	4	2	3	1	0.8	0.4
7	*N*	0.53	4	3	2	1	0.8	0.6
8	*N*	0.51	4	4	1	1	0.8	0.8
9	*P*	0.50	5	4	0	1	1.0	0.8
10	*N*	0.40	5	5	0	0	1.0	1.0



Common Issues in Classifiers

Work Integrated Learning Programmes

Sigmoid Function

- The main class of interest is **rare**.
- The data set distribution reflects a significant majority of the negative class (Eg., Job Offered = Yes/1) and a minority positive class (Eg., Job Offered = No/0)

- For Another Example,
  - fraud detection applications, the class of interest (or positive class) is *"fraud,"*
  - *m*edical tests, there may be a rare class, such as *"cancer"*

- **Accuracy might not be a good option** for measuring performance in case of class imbalance problem

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Solution to Class Imbalance

- Generate Synthetic Samples
- New samples based on the distances between the point and its nearest neighbors E.g. Synthetic Minority Oversampling Technique, or SMOTE class in sklearn
- Change the performance metric : Use Recall, Precision or ROC curves instead of accuracy
- Try different algorithms : Some algorithms as Support Vector Machines and Tree-Based algorithms may work better with imbalanced classes. We will discuss these post mid term

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Types of Classification Based on the Output Labels

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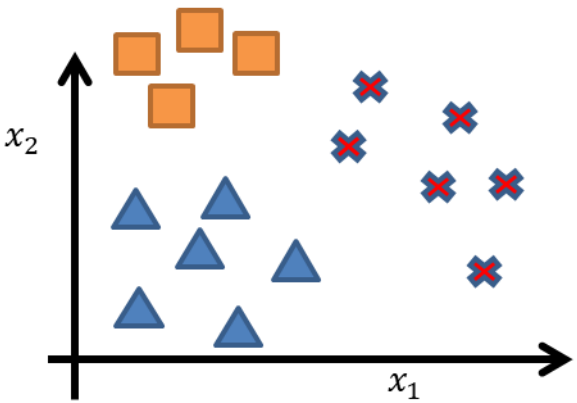
Types of Classification

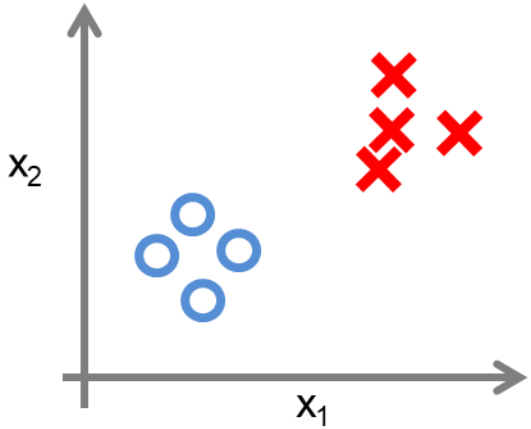
Output Labels

- Target Concept

Multi Class

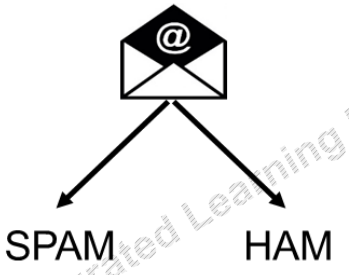


Binary



Enjoy Sports

Spam Classifier



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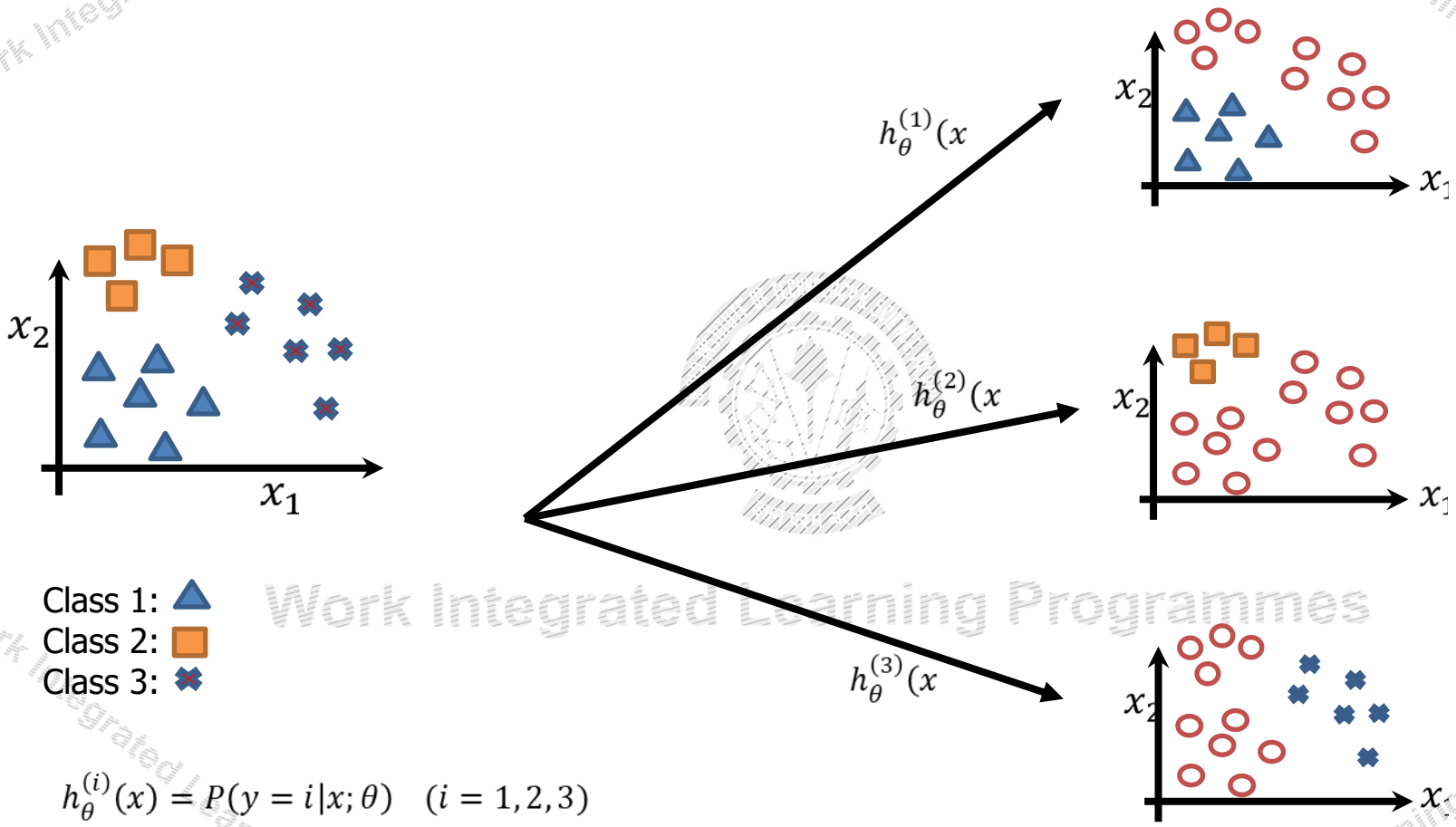
Prediction – Multi class Classification

One Vs All Strategy(one-vs-rest)



Class 1: ▲
Class 2: ■
Class 3: ✖

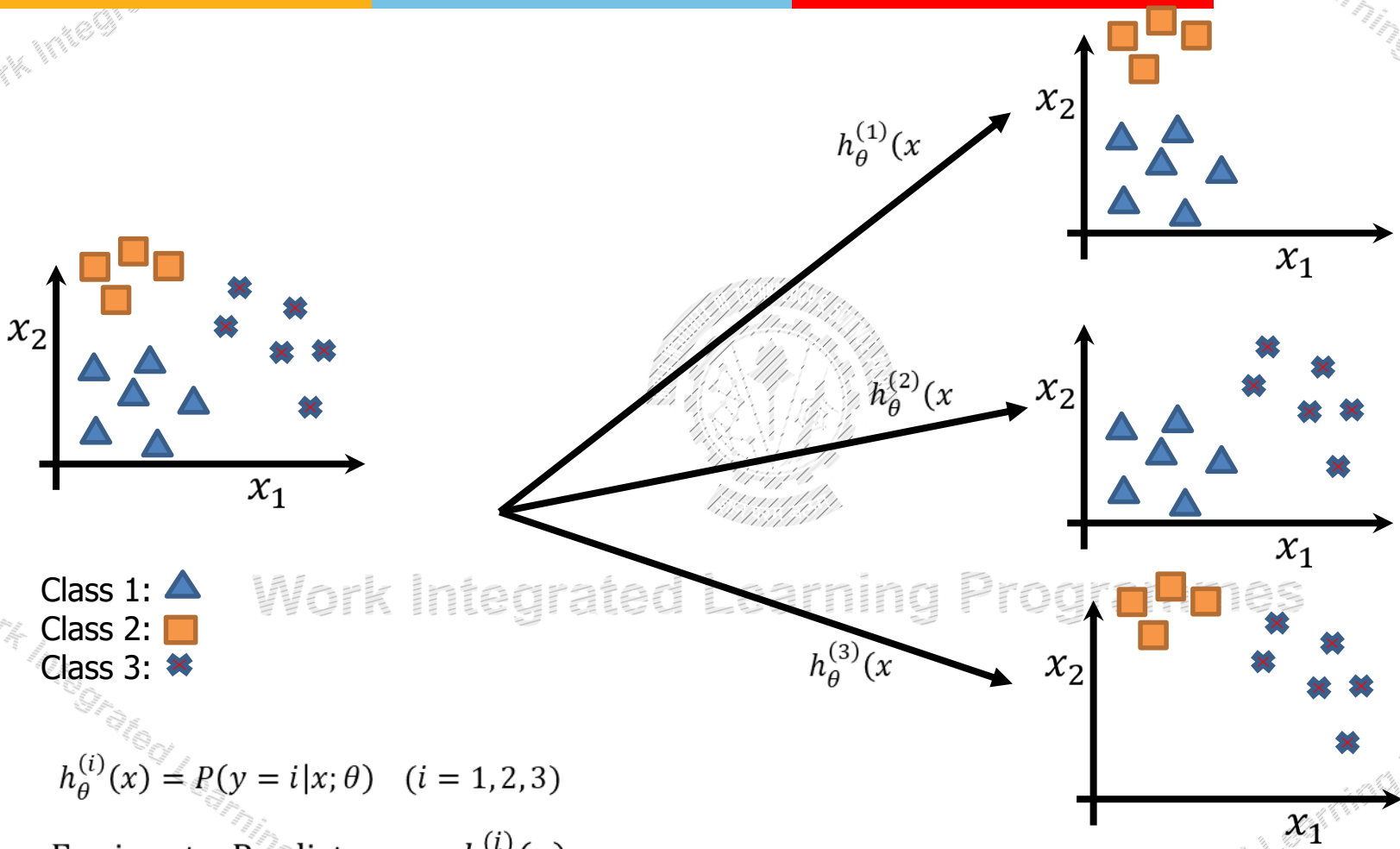
$$h_\theta^{(i)}(x) = P(y = i|x;\theta) \quad (i = 1,2,3)$$

$$\text{For input x Predict}_{\text{i}} : \ \max h_\theta^{(i)}(x)$$

Note: Scikit-Learn detects when you try to use a binary classification algorithm for a multi-class classification task, and it automatically runs OvA (except for SVM classifiers for which it uses OvO)

Prediction – Multi class Classification

One Vs One Strategy



Class 1: ▲
Class 2: ■
Class 3: ✖

$$h_\theta^{(i)}(x) = P(y = i|x;\theta) \quad (i = 1,2,3)$$

$$\text{For input x Predict}_{\text{i}} : \ \max h_\theta^{(i)}(x)$$

N × (N – 1) / 2 classifiers

Logistic regression (Classification)- Summary

- **Model**

$$h_\theta(x) = P(Y = 1|X_1, X_2, \cdots, X_n) = \frac{1}{1+e^{-\theta^\top x}}$$

- **Cost function**

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \mathrm{Cost}(h_\theta(x^{(i)}), y^{(i)})) \qquad \mathrm{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

- **Learning**

Gradient descent: Repeat $\{\theta_j := \theta_j - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_j^{(i)}\}$

- **Inference**

$$\hat{Y} = h_\theta(x^{\text{test}}) = \frac{1}{1 + e^{-\theta^\top x^{\text{test}}}}$$

Note:

- σ(t) < 0.5 when t < 0, and σ(t) ≥ 0.5 when t ≥ 0, so a Logistic model predicts 1 if xTθ is positive, and 0 if it is negative
- logit(p) = log(p / (1 - p)), is the inverse of the logistic function. Indeed, if you compute the logit of the estimated probability p, you will find that the result is t. The logit is also called the log-odds